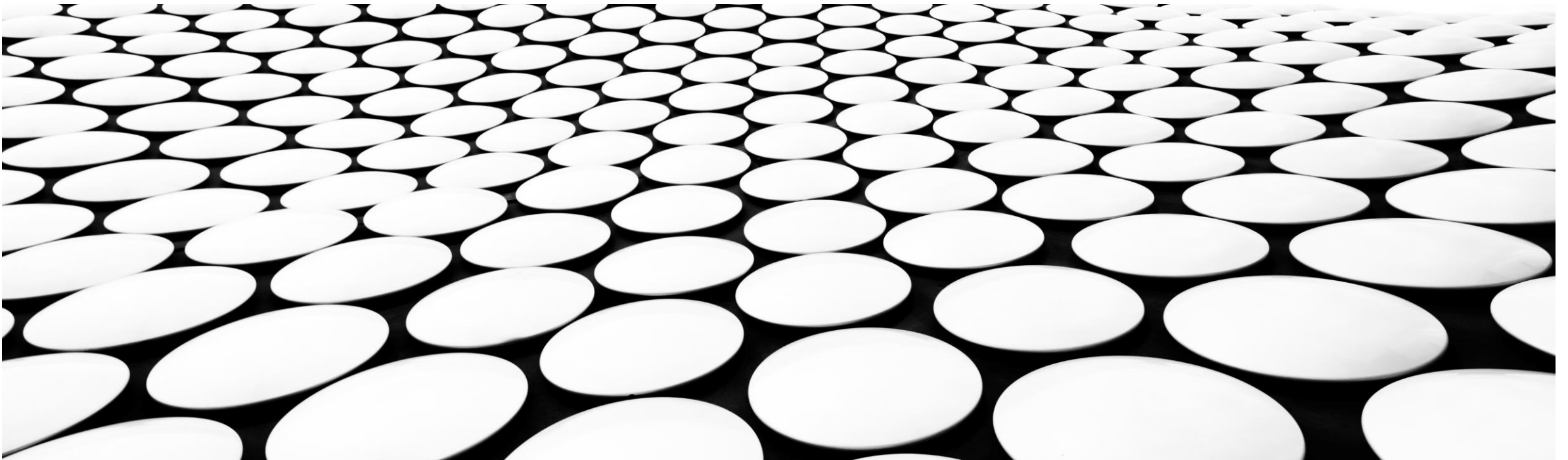

EF2452

MATHEMATICS FOR ECONOMICS AND FINANCE

Lecture 1: Mathematical Terms And Notations

DR LEE PAK YEE



Diagrams, graphs and selected examples are adapted from the course textbook (Anthony & Biggs 2024). Copyright remains with the publisher.

WHY MATHEMATICS FOR ECONOMICS?

- Mathematics is an important language of economics
- It is a useful tool to study economics at all different levels.
- Economic theories simplify the real world.
- Mathematics helps express economic relationships in a manageable form.

ELEMENTS OF A MATHEMATICAL MODEL

- Variables: Real Numbers
 - For example: Price (p), Quantity (q), Inputs (K, L)
- Relationships: Functions
 - For example: Demand Function: $q^D(p) = a - bp$



MATHEMATICAL TERMS AND NOTATIONS

- Before studying mathematical concepts, we need to understand some basic mathematical terms and notations.
- These terms and notations will be used throughout the course and other courses.
- We now begin with sets.

SETS

- A **Set** is a collection of objects.
- The objects in a set are called **elements** (or members) of the set.
- Sets are usually denoted by capital letters, such as A, B, S, D
- The elements of a set are enclosed in curly brackets $\{ \}$.

SETS

- One way of specifying a set is by enumeration, listing all the elements of the set in any order.

$$A = \{1, 2, 4, 6, 7, 9\}$$

$$Z = \{1, 2, 3, 4\}$$

SETS

- If an element x is in a **set** A (x is a member of A or x belongs to A) , we write

$$x \in A$$

- If x is not in A , we write

$$x \notin A$$

SETS

EXAMPLE

$$X = \{2, 3, 5, 7, 8, 9\} \text{ and } Y = \{1, 4, 5, 7, 9\}$$

We say that

$$2 \in X$$

$$3 \notin Y$$

SETS

- Another way of specifying a set is by description, describing a property that identifies which objects belong to the set

$$B = \{x | x \text{ is an even integer}\}$$

$$Z = \{n | n \text{ is a positive whole number less than } 5\}$$

SETS

- A is a **finite set** if it has a finite number of elements.

$$A = \{1, 2, 4, 6, 7, 9\}$$

- B is an **infinite set** if it has an infinite number of elements.

$$B = \{x | x \text{ is an even integer}\}$$

SETS

- We define the **empty set**, $\emptyset$ as the set which has no elements (no members)

SETS

- If every member of A is also a member of B (A is a **subset** of B), we write

$$A \subseteq B \text{ or } B \supseteq A$$

- If $A \subseteq B$, but at least one member of B is not a member of A (A is a **proper subset** of B), we write

$$A \subset B$$

SETS

- Two sets A and B are said to be equal ($A = B$) if

$$A \subseteq B \text{ and } A \supseteq B$$

SETS

EXAMPLE

$$A = \{1, 2, 3\}, B = \{3, 1, 2\} \text{ and } C = \{1, 2, 3, 4, 5, 6\}$$

We say that

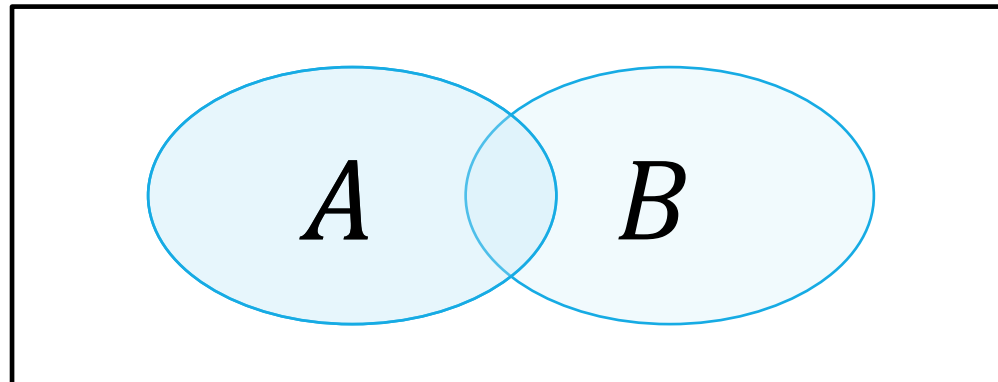
$$A \subset C$$

$$A = B$$

SETS

- The **union** of two sets A and B is the set containing all members that belong to A , B or both (A and B):

$$A \cup B = \{x | x \in A \text{ or } x \in B\}$$



SETS

EXAMPLE

$$X = \{2, 3, 5, 7, 8, 9\}, Y = \{1, 4, 5, 7, 9\} \text{ and } Z = \{1, 2, 3, 4\}$$

$$X \cup Y = \{1, 2, 3, 4, 5, 7, 8, 9\}$$

$$Y \cup Z = \{1, 2, 3, 4, 5, 7, 9\}$$

SETS

EXAMPLE

$$X = \{2, 3, 5, 7, 8, 9\}, Y = \{1, 4, 5, 7, 9\} \text{ and } Z = \{1, 2, 3, 4\}$$

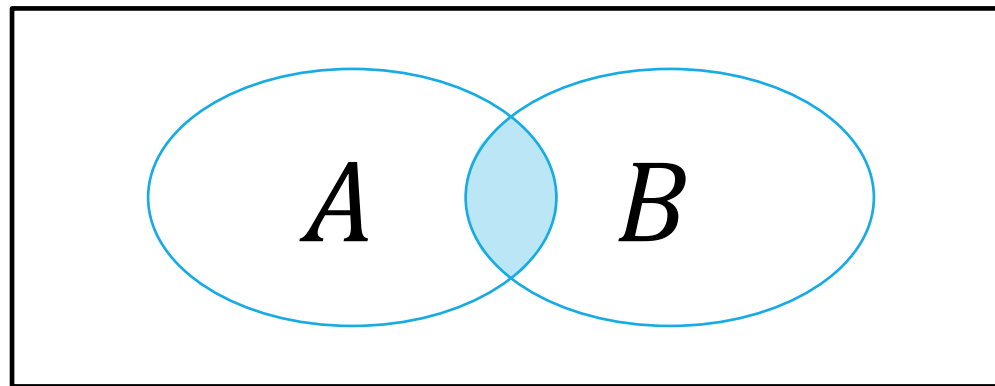
$$X \cup Y = \{1, 2, 3, 4, 5, 7, 8, 9\}$$

$$Z \subseteq X \cup Y$$

SETS

- The **intersection** of two sets A and B is the set containing all members that belong to both A and B :

$$A \cap B = \{x | x \in A \text{ and } x \in B\}$$



SETS

EXAMPLE

$$X = \{2, 3, 5, 7, 8, 9\}, Y = \{1, 4, 5, 7, 9\} \text{ and } Z = \{1, 2, 3, 4\}$$

$$X \cap Y = \{5, 7, 9\}$$

$$(X \cap Y) \cap Z = \emptyset$$

SETS

EXAMPLE

$$X = \{2, 3, 5, 7, 8, 9\}, Y = \{1, 4, 5, 7, 9\} \text{ and } Z = \{1, 2, 3, 4\}$$

$$Y \cap Z = \{1, 4\}$$

$$X \cap (Y \cap Z) = \emptyset$$

Microeconomics: Consumer Theory

- The consumer's budget set is the set of all affordable bundles;

$$B = \{(x_1, x_2, \dots, x_n \in X) | p_1 x_1 + p_2 x_2 + \dots + p_n x_n \leq m\}$$

Game Theory/ Industrial Organization

- A strategic game consists of:
 - A finite set of players: $\{1, 2, 3, \dots, N\}$
 - A set of strategies (actions) for each player

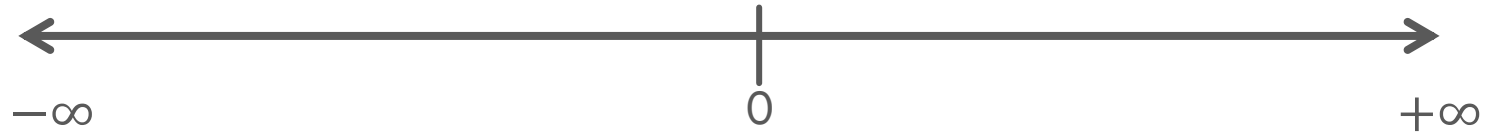
$$s_i \in S_i$$

$$S = (S_1, S_2, \dots, S_N)$$

- A set of payoff functions which give players' utility for each profile of strategies.

NUMBERS

- A **real number** $\mathbb{R}$ is any number along a continuous line (real line, $\mathbb{R}^1$).



- The set of non-negative numbers is denoted by $\mathbb{R}_+$ ($\mathbb{R}_+ \subset \mathbb{R}$) and it contains all real numbers greater than or equal to zero.

$$\mathbb{R}_+ = \{x | x \geq 0\}$$

NUMBERS

- The set of **integers** (numbers that can be expressed without a fraction or decimal) is denoted by $\mathbb{Z}$.

$$\mathbb{Z} = \{ \dots, -3, -2, -1, 0, 1, 2, 3 \dots \}$$

- The set of **natural numbers** (positive integers) is denoted by $\mathbb{N}$.

$$\mathbb{N} = \{ 1, 2, 3 \dots \}$$

ORDERED PAIRS

- An **ordered pair** is a pair of elements with a particular relation between them. In general,

$$(x, y) \neq (y, x)$$

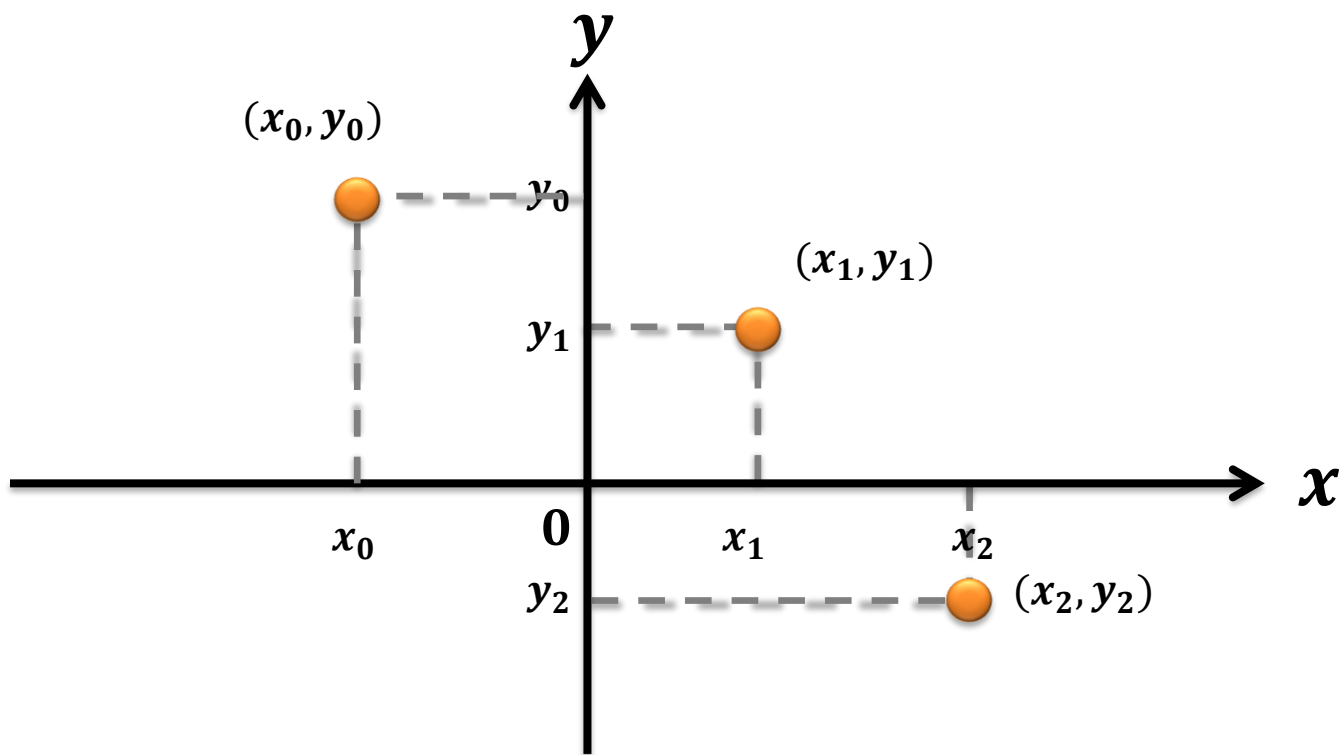
- For example:

$$\mathbb{R}_{++}^2 = \mathbb{R}_{++} \times \mathbb{R}_{++} = \{(x, y) \in \mathbb{R}^2 \mid x > 0, y > 0\}$$

It is also known as the first quadrant.

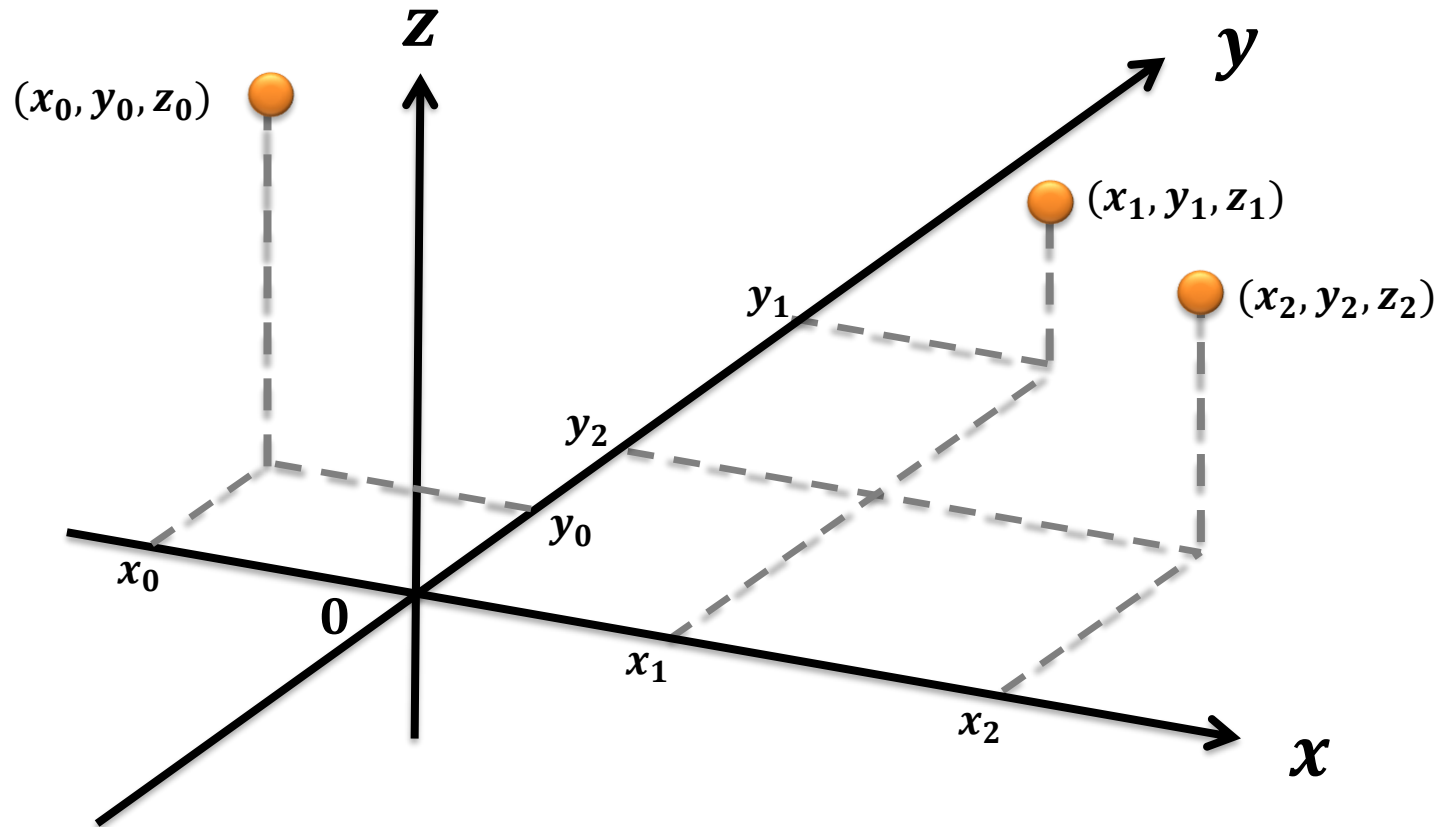
ORDERED PAIRS

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(x, y) \mid x \in \mathbb{R}, y \in \mathbb{R}\}$$



ORDERED PAIRS

$$\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R} = \{(x, y, z) | x \in \mathbb{R}, y \in \mathbb{R}, z \in \mathbb{R}\}$$



ORDER PAIRS

EXAMPLE

$$A = \{1, 2\} \text{ and } B = \{a, b, c\}$$

Then

$$A \times B = \{(x, y) | x \in A, y \in B\}$$

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$$

Microeconomics: Supply and Demand

- A demand schedule can be represented as a set of ordered pairs in $\mathbb{R}^2$:

$$D = \{(q_1, p_1), (q_2, p_2), \dots, (q_n, p_n)\}$$

where p_i is the price and q_i is the corresponding quantity demanded.

Demand as a Set

- A demand set contains all ordered pairs (q, p) in $\mathbb{R}^2$ that satisfy the demand relationship

$$D = \{(q, p) | 6q + 8p = 125\}$$

where p is the price and q is the corresponding quantity demanded.

- The demand set D contains the point $(7.5, 10)$. This means that when the price is 10, the quantity demanded is 7.5.

From a Demand Set to a Demand Function

$$D = \{(q, p) | 6q + 8p = 125\}$$

Simply rearrange for q

$$q = \frac{125}{6} - \frac{8p}{6}$$

- Therefore, the demand function is

$$q^D(p) = \frac{125}{6} - \frac{4p}{3}$$

Supply as a Set

- A supply set contains all ordered pairs (q, p) in $\mathbb{R}^2$ that satisfy the supply relationship

$$S = \{(q, p) | 2q - 5p = -12\}$$

where p is the price and q is the corresponding quantity supplied.

- Therefore, the supply function is

$$q^s(p) = \frac{5p}{2} - 6$$

Market Equilibrium

- The ordered pair (q^*, p^*) which is in the equilibrium set $E = S \cap D$ must be in both S and D . Thus (q^*, p^*) must satisfy the two equations.

$$q^S(p) = \frac{5p}{2} - 6$$

$$q^D(p) = \frac{125}{6} - \frac{4p}{3}$$

Solving the above equations gives $q^* = 11.5$ and $p^* = 7$. In other words, the equilibrium set E contains the ordered pair $(11.5, 7)$.

FUNCTIONS

- Let A and B be arbitrary sets. A function f from A to B assigns exactly one member of B to each member of A .
- A is the **domain** and B is the **codomain**.
- The **range** (or **image**) of a function is the set of all values actually taken by the function.

FUNCTIONS



- For example,

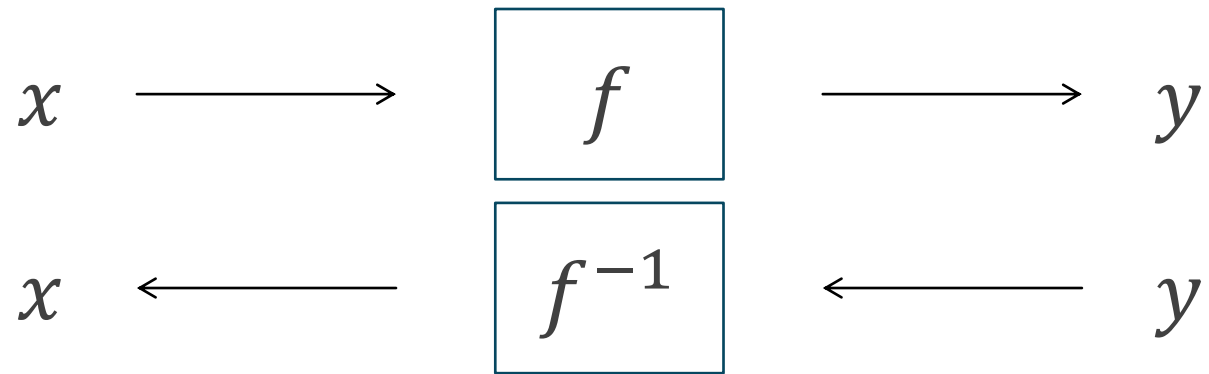
$f(x) = 2x$ is the same as

$$f(y) = 2y \text{ or } f(t) = 2t$$

- The variable, x used in a function is a **dummy variable**. Its name does not change the function.

FUNCTIONS

- An **inverse function** reverses the input and output.



FUNCTIONS

- Suppose

$$f(x) = 2x + 7$$

$$y = 2x + 7$$

$$2x = y - 7$$

$$x = (y - 7)/2$$

$$f^{-1}(y) = (y - 7)/2$$

FUNCTIONS

- If f and g are inverses of each other, it implies

$$y = f(x)$$

if and only if

$$g(y) = x$$

$$f = g^{-1}$$

.

FUNCTIONS

- A **composite function** applies one function after another.

$$(s \circ r)(x) = s(r(x))$$

- The order of the function matters. In general,

$$(s \circ r) \neq (r \circ s)$$

FUNCTIONS

- Suppose that

$$r(x) = x^2 + 2$$

$$s(x) = x^3$$

Then the composite function

$$k(x) = s \circ r = s(r(x)) = s(x^2 + 2) = (x^2 + 2)^3$$

FUNCTIONS

- An **identity function** maps each value to itself.

$$i(x) = x$$

- The composition of f and g is the identity function.

$$g \circ f = f^{-1} \circ f = i$$

▪

FUNCTIONS

- Suppose that

$$f(x) = 2x \text{ and } f^{-1} = g(x) = \frac{x}{2}$$

Then the composite function

$$f^{-1} \circ f = (g \circ f)(x) = g(f(x)) = g(2x) = \frac{2x}{2} = x = i(x)$$

The composite function $f \circ f^{-1}$ is also equal to $i(x)$.

FUNCTIONS

- A **quadratic equation** has the form

$$ax^2 + bx + c = 0,$$

where $a \neq 0$

- Its solutions are the values of x that satisfy the equation.

FUNCTIONS

- In some cases, the **quadratic equation** can be factorized;

$$x^2 - 5x + 6 = 0$$

After factorizing

$$(x - 2)(x - 3) = 0$$

Solution: 2 roots are

$$x_1 = 2 \text{ and } x_2 = 3$$

FUNCTIONS

- Quadratic equations can be solved using the quadratic formula

$$x_1, x_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Since $a \neq 0$

$$ax^2 + bx + c = 0$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

FUNCTIONS

By using completing the square

$$\left(x + \frac{b}{2a}\right)^2 = x^2 + 2x\left(\frac{b}{2a}\right) + \left(\frac{b}{2a}\right)^2 = x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}$$

$$x^2 + \frac{b}{a}x + \frac{c}{a} = \left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2} + \frac{c}{a} = 0$$

FUNCTIONS

$$x^2 + \frac{b}{a}x + \frac{c}{a} = \left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a^2} + \frac{c}{a} = 0$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a}$$

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

FUNCTIONS

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

$$x + \frac{b}{2a} = \frac{\sqrt{b^2 - 4ac}}{2a} \text{ or } x + \frac{b}{2a} = -\frac{\sqrt{b^2 - 4ac}}{2a}$$

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ or } x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

FUNCTIONS

Then:

- if $b^2 - 4ac < 0$, the equation has no real solutions;
- if $b^2 - 4ac = 0$, the equation has exactly one solution, $x = -\frac{b}{2a}$;
- if $b^2 - 4ac > 0$, the equation has two solutions: $x_1, x_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$;

FUNCTIONS

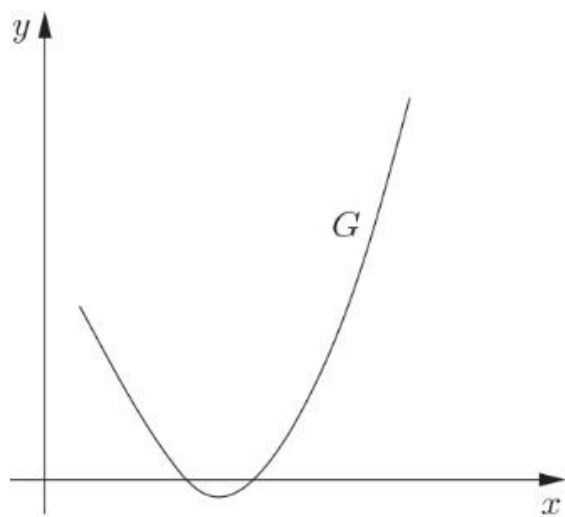
- We can also apply set concepts to study functions.
- Suppose that

$$G = \{(x, y) \mid y = x^2 - 5x + 6\}$$

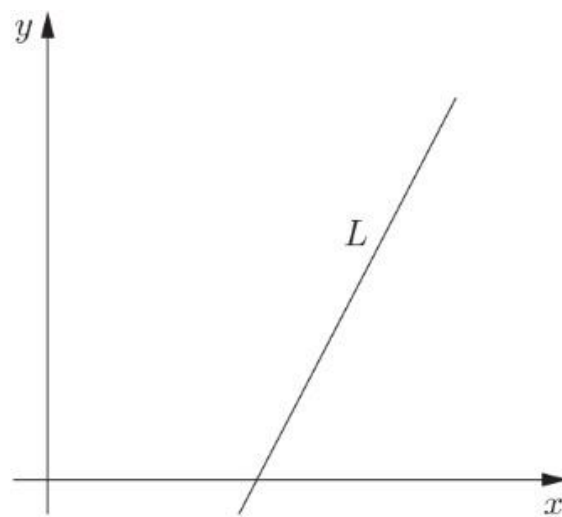
$$L = \{(x, y) \mid y = 2x - 6\}$$

The intersection set $G \cap L$ contains all ordered pairs that satisfy both equations.

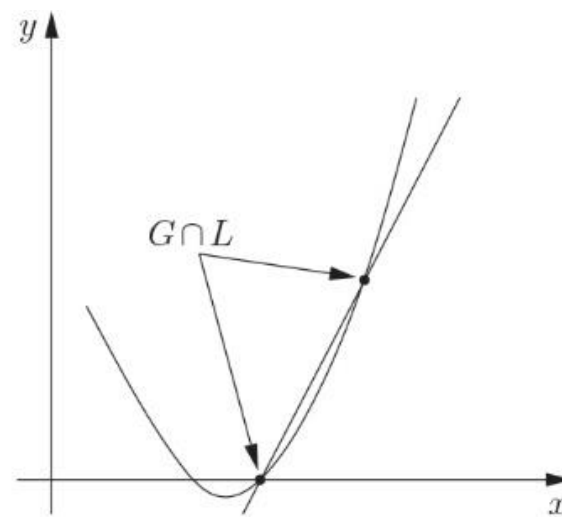
FUNCTIONS



(a)



(b)



(c)

FUNCTIONS

- By eliminating y

$$x^2 - 5x + 6 = 2x - 6$$

$$x^2 - 7x + 12 = 0$$

- After factorization:

$$(x - 3)(x - 4) = 0$$

The solutions are $x_1 = 3$ and $x_2 = 4$. Substituting these values into $y = 2x - 6$ gives the ordered pairs $(3, 0)$ and $(4, 2)$. Therefore, $G \cap L = \{(3, 0), (4, 2)\}$.



ANY QUESTION?